

DYNAMIC INSULATED SHUTTER SYSTEM FOR A DOUBLE-LAYER BUILDING FAÇADE: A CASE STUDY WITH PHYSICAL PROTOTYPE

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ABSTRACT

Adaptive building envelopes (ABEs) enable the dynamic response of building enclosures to environmental conditions. Movable window insulation is a powerful adaptive technique, but its potential has not been sufficiently explored. Based on a case study, an insulated shutter system with a double-layer façade capable of adapting to varying environmental conditions, especially the hot days and cold nights of the desert, was investigated. Specifically, the control operation schedules of different facades and layers were based on temperature and solar position to maximize energy efficiency, daylighting, and thermal comfort. Dynamic shading simulations were used to optimize the façade performance. Subsequently, temperature thresholds are established based on the optimization of vital parameters like Useful Daylight Illuminance (UDI), Energy Use Intensity (EUI), and Predicted Mean Value (PMV). Moreover, two complementary physical models were tested to evaluate natural ventilation and the physical mechanism of operation respectively.

Keywords: building energy simulation, adaptive building envelope, movable window insulation, control logic, operation schedule.

1 INTRODUCTION

In the US, the building sector consumes about 40% of national energy and uses more than 70% of electricity [1]. Building envelope systems are essential for managing buildings' heating and cooling loads, as over 40% of building energy is dedicated to balancing the heat losses and gains at building envelopes [2]. Therefore, improving building envelopes is a practical solution to conserve energy and contribute to decarbonization [3]. As an alternative to a passive system with a static envelope, adaptive building envelopes (ABEs) can change their states and functions over time in response to varying ambient conditions, acting as climate mediators that negotiate between multiple competing priorities like comfort needs, views, and energy use [4]. Movable window insulation was chosen as the prototype of this research project due to its significant impacts on multiple physical domains, relatively simple structure, considerable energy-saving potential, and lack of systematic studies [5].

An investigation into the cost-effectiveness of moveable window insulation in the US [5] has already shown its significant potential to conserve energy and improve building performance with considerable financial payback in hot or cold climates compared with other window protection measures. Two key features of a movable window insulation system are the configuration and embedded control logic. In that study, the sliding window insulation was attached to typical mid-size windows embedded in opaque walls, controlled independently on different facades, and the control algorithm was based on complicated heat balance calculations used in EnergyPlus. Another study looks into the benefits of Switchable Insulation Systems (SIS) with rotatable insulation layers applied to residential buildings in France, achieving up to 63% annual

energy savings for the whole building and 82% for one unit [6]. Similarly, the control logic is based on various measurements like temperatures at multiple spots to determine the operation mode.

The articles mentioned above demonstrate that movable window insulation and its variants have great potential to improve building performance, but there are still plenty of gaps and disadvantages to be addressed: 1) There is no daylight when insulation is closed; 2) There is no natural ventilation when insulation is closed; 3) Movable window insulation is normally attached to windows as an “accessory”, not an integrated part of the architecture; 4) Control logic can become so complex that it is challenging to implement and to be understood by users and architects; 5) Windows on different facades are controlled separately without whole-building control approaches; 6) Lack of holistic consideration to balance energy consumption and building performance (human comfort), as energy reduction is often prioritized. Given the above research gaps, this study begins with the architects’ holistic perspective and introduces a dynamic insulated shutter system to tackle those problems.

The shutter system serves as a well-developed case of a smart movable window insulation system. The term “smart” here denotes a system that can react to varying conditions, which can be substituted with other terms such as dynamic, responsive, or kinetic. As a supplement to studies on movable window insulation that include a universal control algorithm for every window [5], this paper introduces a straightforward building-scale approach with three unfixed temperature thresholds that can be adjusted based on specific climatic conditions. The temperature thresholds are determined through simulations and optimizations. Instead of relying on complicated thermal calculations to determine the operation, the control logic here is simply based on a simple rule: whether the house needs direct sunlight or not. The envelope only responds to temperature (outdoor and indoor) and solar position. Simulation results such as energy efficiency (EUI), daylighting (UDI), and thermal comfort (PMV & comfort temperature range) are analyzed and optimized. Firstly, the study aims to demonstrate that the system can contribute to energy conservation and enhanced building performance with adequate daylight and natural ventilation, even under severe climatic conditions. Beyond that, it focuses on the architectural level instead of engineering or mathematical levels, introducing a simple whole-building control logic to handle complex environmental conditions. Ultimately, it tries to achieve a subtle balance among several expectations, and the final optimization logic can be customized based on the user’s requirements. The dynamic insulated shutter system is rooted in a case study of a small desert house in Phoenix designed by the authors, but the methods developed in this study are expected to be generally applicable.

2 METHODS

2.1 Case Study

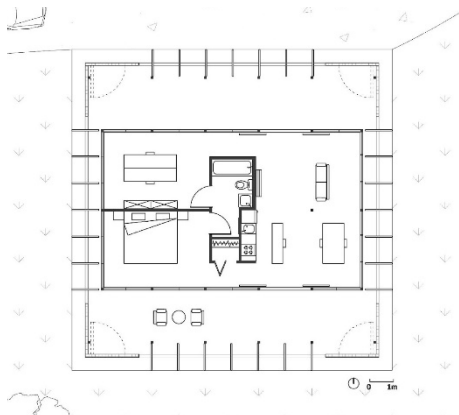


Figure 1: Floor plan.



Figure 1: Perspective view of the house.

The project is situated in a desert area at Taliesin West, on the east of Phoenix. The small house is designed for a couple and is not connected to the power or water grid, requiring basic living standards to be met

through architectural solutions. The primary concern in this context is addressing heat, but cold winter weather is still a concern, so a holistic approach is needed for every season. Figure 1 shows the floor plan of the house. It is a 12m-by-12m pre-fabricated house with a shell of insulation layer and an inner glazing layer. There are two semi-outdoor spaces on the north and south sides, the north one mainly serves as an entrance porch and the south one provides a winter sitting place with sufficient daylight. The living room, bedroom, and study are arranged around the core, which contains a kitchen, a wardrobe, and a bathroom.

The main research objectives of the paper are the facades of the house, which consist of rotatable insulated shutters and fixed translucent aerogel panels. The shutters are constructed with two panes of 10mm thick wood and a 30mm vacuum insulation panel (VIP) in between, resulting in 50mm thick panels that have R-values of 30 m²/kW. While VIP offers high thermal performance, its adoption is limited due to the potential for damage during installation and operation periods [7]. In this study, VIP was chosen to emphasize the effects of the facades' different states. Pivots were embedded in each of the shutters to enable rotation without compromising insulation effectiveness. To prevent air leakage, shutters are overlapped with each other, so the facades are not completely flat when closed. On the other hand, translucent panels share the same 50mm thickness and R-value of 30 Km²/W as the shutters, but they are designed with two glass panes filled with aerogel in between. Aerogel, an effective insulation material, exhibits a much lower thermal conductivity than conventional materials like mineral wool [8]. Additionally, it allows high transmittance to provide natural daylight, illuminating the interior even when all shutters are closed. Furthermore, the house features four translucent doors at each corner, constructed with the same materials as the aerogel panels, serving as entrances or exits. This design offers residents flexibility to enter or leave the house in any direction, regardless of contextual constraints. Overall, each facade of the house maintains a uniform appearance, and the entire envelope is symmetrical.

2.2 Control Logic of Dynamic Shutters

The control logic of the shutters is based on two factors: temperature and solar position (time). Time here stands in for solar position as it is easier to implement in simulation. The control logic is shown in Figure 4; in short, there are three categories: needing direct sun, no direct sun, and fully closed for thermal protection. The decision-making level is purely based on outdoor temperature and applies to all facades. Within each category, the states of shutters on different facades depend on the position of the sun.

The interior daylighting condition is acceptable in this study when shutters are all closed, meaning that thermal comfort is the main issue to be addressed here, and daylighting is a secondary variable to be considered. In addition, when outside is too hot or cold, all shutters should be closed to enhance thermal comfort and reduce energy loads. In the simulation, this means that if the hourly dry-bulb temperature is lower than T_1 or higher than T_2 (Figure 4), all shutters will be closed. It is noted that closed shutters only stop natural ventilation, while mechanical ventilation still works to provide fresh air.

Meanwhile, when it is cool outside and direct sunlight is needed, the house opens shutters when the dry-bulb temperatures are between T_1 and T_2 . The south façade is always open during the day as the sun is low and mainly comes from the south on cool days, the east façade opens in the morning and the west façade opens in the afternoon, all shutters are closed at night. In the simulation, the setting is to close all shutters when there is no direct solar radiation, open the south façade as long as there is radiation, open the west shutters after 2 p.m. with solar radiation, and open the east shutters before 10 a.m. with solar radiation. In those states, limited shutters are open most of the time because insulation is still important to maintain indoor comfort when the outside is cool.

When the outdoor environment is not too hot, indicating that the outdoor dry-bulb temperatures fall between T_2 and T_3 , direct sunlight is unwanted. In this regard, the insulation is not crucial. The north façade is always open to provide diffused daylight, the west façade opens in the morning and the east façade opens in the afternoon. At night, the north, east, and west facades will always open for potential night flushing opportunities. In the simulation, the north façade is always open, regardless of day or night; the west opens after noon and the east opens before noon, while the south is always closed. In this scenario with warm

outdoor temperatures, natural ventilation is frequently needed to cool down the indoor temperature. The sliding windows are supposed to be opened when the outdoor temperature is lower than the indoor temperature and the indoor temperature is at least 22°C. Those ventilation settings are realized in simulation by using “Honeybee Ventilation Control” and “Honeybee Window Opening” components. Honeybee is a grasshopper plug-in that supports daylighting and thermodynamic modeling using OpenStudio, and the components can control the ventilation state of operable windows based on point-in-time temperatures and temperature thresholds designed for natural ventilation. The prevailing wind in Taliesin West mainly comes from the southwest, and natural ventilation may have a considerable impact in regulating indoor temperature as the house is relatively small, even though the central core may block most cross ventilation.

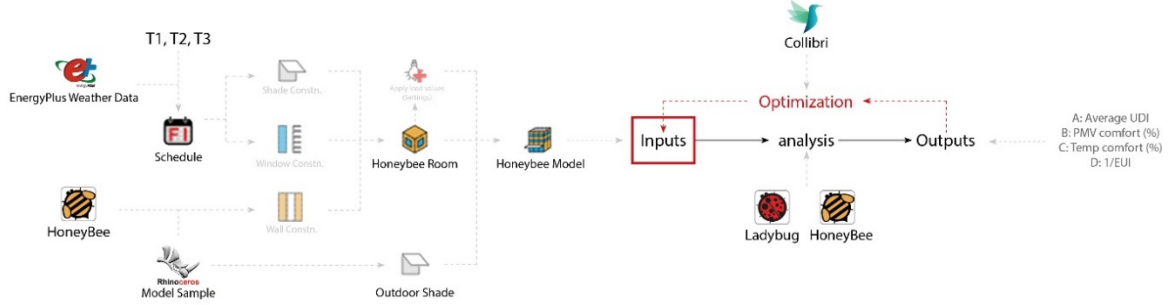


Figure 2: Simulation workflow overview.

It should be mentioned that the control logic presented here (Figure 3) is just one of several acceptable strategies, instead of a perfect solution. There are other possibilities with different details as well, for instance, between T_1 and T_2 the south façade can be closed when sunlight is not directly from the south, or between T_2 and T_3 all facades can be open at night to maximize the night flushing effect. Besides, other factors like views, privacy, residents’ habits, and distinguished daylighting requirements for different spaces in the house can also be considered. Therefore, the solution here is an exemplified simple proposal to be analyzed, not sophisticated or the best among all options. However, the key principle of different potential logics is the same which is already expressed in the proposal: using the outdoor temperature to decide whether direct and diffused sunlight are wanted or not. The reason why indoor temperature is not used here as the decision-maker is that outdoor dry-bulb temperature can be directly extracted from the climate file before simulation, while indoor temperature can only emerge after the simulation is completed. In prototypes or real projects, point-in-time indoor temperature might be a better input. Compared with other existing control logic of movable window insulation or adaptive building envelopes [9], the control logic presented here has two main distinctions: unfixed temperature thresholds and results optimization.

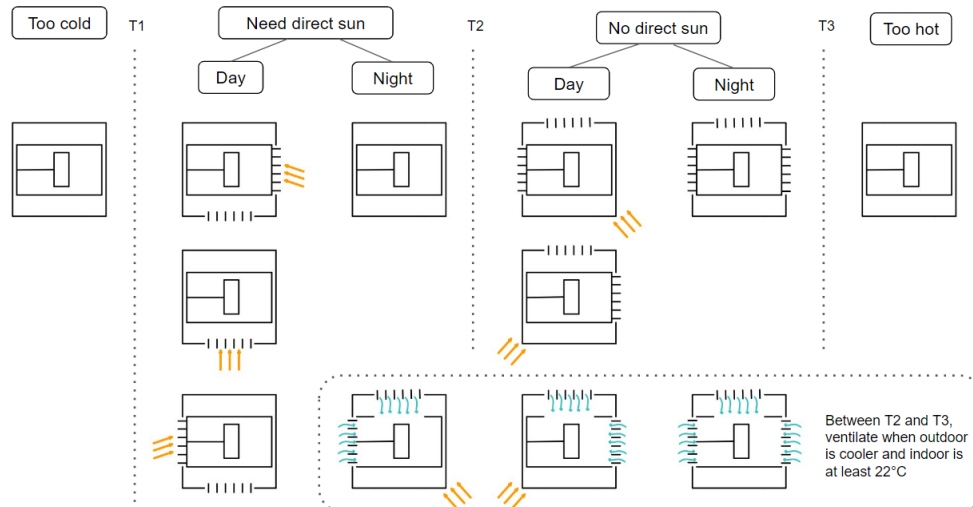


Figure 3: Control logic of the dynamic shutters (T1, T2, T3 represent temperatures).

2.3 Physical Prototype

In the development of the physical prototype, adjustments to its scale were made relative to the original design. These modifications, guided by considerations of practicality and feasibility, led to a minimization of its components while maximizing the thermal properties simulation of the scaled model. The objective was to create a streamlined prototype, primarily customizing for effective hotbox simulation. The hotbox is an enclosed container using strong light illuminance to simulate heat, so the thermal performance of the physical model can be examined. As a preliminary exploration, in the prototype, only one face is designed as a responsive envelope, incorporating three shutters as the simplified version of the original design.

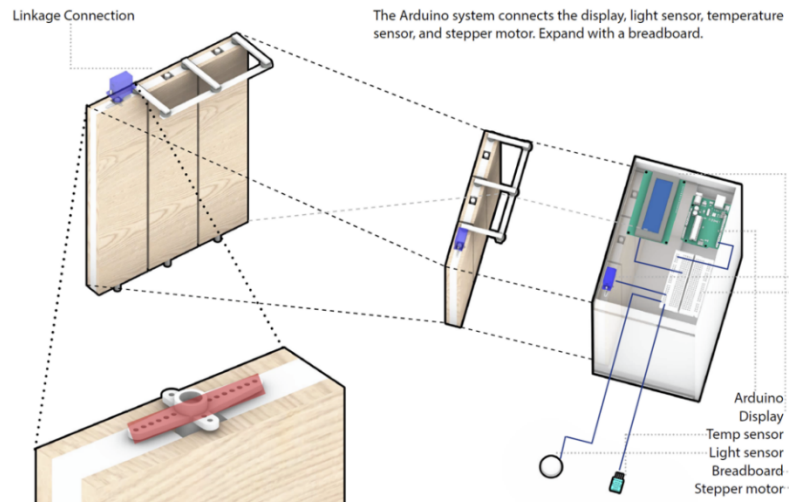


Figure 4: Physical Prototype Setting and Details.

The control system of the prototype employs an Arduino setup (Arduino UNO R3), orchestrating the coordination of various components: an I2C IIC 1602 LCD Display for output visualization, a BH1750fvi light sensor, a DHT11 temperature sensor, and Micro Servos SG90 stepper motors. The operational logic of this system assesses environmental conditions using data from the temperature and light sensors. This data then informs the stepper motors to adjust the shutters' aperture, while a linkage system controls the opening and closing of all the shutters, altering their open or closed state. Real-time data is continuously displayed on the LCD screen, providing immediate feedback on environmental interactions. This approach ensures that the prototype not only simulates the intended functionality but also allows for adjustments and improvements based on immediate data feedback.

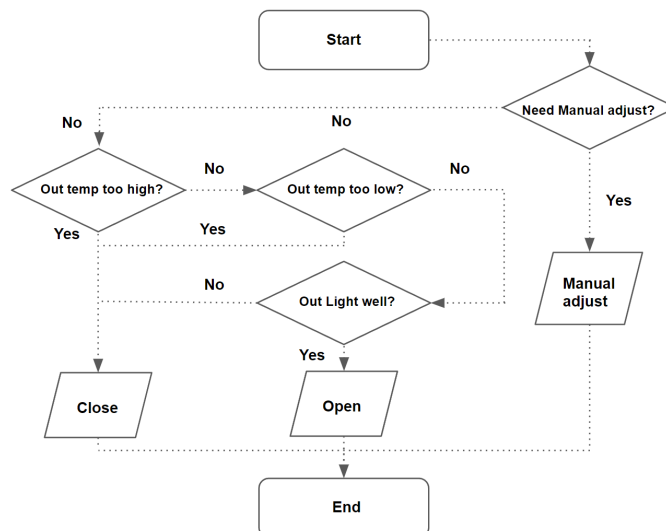


Figure 5: Workflow of Control.

As delineated in the control logic framework, a precise control workflow is established (refer to Figure 5). This workflow distinguishes itself by integrating holistic control logic with detailed, real-time decision-making processes. The primary criterion in this workflow is temperature regulation. If temperatures fall outside the designated range (T_1 to T_3 , which denotes the optimal temperature spectrum), the building envelope is programmed to close, thereby safeguarding the indoor environment. Within the acceptable temperature range (between T_1 and T_3), the decision to open or close the shutters hinges on lighting conditions. The parameter set for acceptable illuminance is 100-3000 Lux, correlating with the definition of Useful Daylight Illuminance (UDI). If illuminance falls within this spectrum, the facade is maintained in an open state. Conversely, if the illumination strays from this range, indicating unsuitable lighting conditions, the louvers are closed. Moreover, the workflow incorporates a provision for manual control to cater to specific human needs, acknowledging that personal preferences and requirements may vary. This aspect of control is prioritized in the decision-making process and, in conjunction with the described workflow, collectively guides the movement logic of the dynamic model. Additionally, simplified hotbox experiments using a scaled prototype are based on this workflow, designed to assess the impact of ventilation on temperature control within the dynamic envelope.

The Hotbox Experiment, a methodological approach for examining the physical properties and performance characteristics of materials or devices, utilizes a specialized sealed and controlled container designed to simulate ideal environmental conditions. This container, equipped with its own heat source—hence the designation "hotbox"—, is often integrated with various sensors to collect data on thermal behavior, ensuring precise manipulation and monitoring of internal conditions to assess material or device performance accurately. Subject to thermal radiation, these materials undergo temperature fluctuations, which are meticulously monitored using sensors integrated into the system. This experimental setup facilitates an in-depth analysis of material and device behavior under various thermal conditions. In the initial experimental configuration, two primary states for the façade were identified: open and closed. The hotbox simulation experiments, conducted with a movable prototype, were pivotal in evaluating the influence of these states on temperature dynamics.

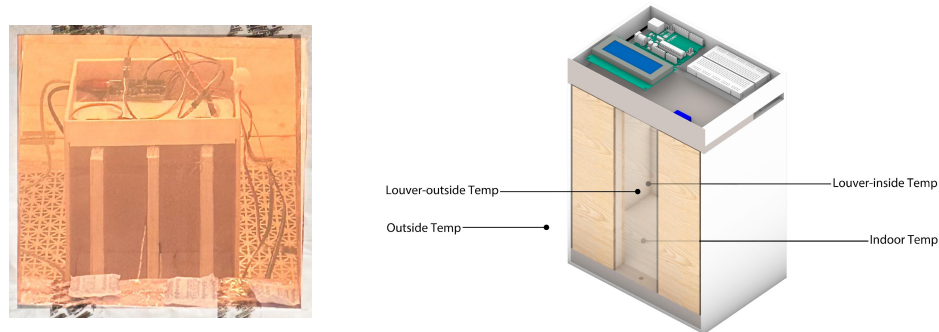


Figure 6: Physical Prototype model and model diagram.

To scale the model properly, the prototype was scaled to a 1:30 ratio of the original 12 x 12m house dimensions, with its envelope's R-value matching the proposed project R-value. This approach facilitated thermal regulation experiments in a controlled environment. Accurate representation of the building's thermal behavior was achieved by meticulous scaling based on R-values—measures of thermal resistance. Higher R-values indicate better insulation. This consideration was crucial for adjusting the prototype's dimensions, particularly wall thickness, to accurately simulate real-world thermal performance. Unlike software simulations that rely on annual data, thus macroscopic in nature, the Hotbox Experiment, or called physical simulation, captures real-time temperature changes, offering a microscopic analysis based on instantaneous data. The hotbox experiments include three states: louvers stationary and fully closed, louvers stationary and fully open, and dynamically adjusted louvers. To simplify and facilitate the experiment, only light control was considered for the dynamic aspect. The experiment had distinct and unalterable stages, ensuring control and clarity. Spanning four hours, the experiment is divided into four one-hour phases, incorporating both turning on and turning off the hotbox light source to simulate day and night temperature

changes and assess the impact of repeated open-close cycles. Four testing points are designated: the indoor side of the louver, the outdoor side of the louver, the Outside, and the Indoor. As depicted in Figure 6, right panel, within the prototype, four testing locations are specified: the indoor side of the louver, the outdoor side of the louver, Outside, and Indoor. Temperature sensors are installed on both sides of the louvers to monitor temperature fluctuations during various state changes. This experimental logic is tested thrice (three states each, totaling nine trials) under three different light source intensities to explore universality and better contrast the temperature variations across states.

Temperature ranges are set to evaluate comfort levels in experiments, focusing on the impact of the dynamic envelope on comfort. The CBE Thermal Comfort Tool [10], employing Operative Temperature and Prevailing mean outdoor temperature, is used for this assessment. The Operative Temperature is defined as the temperature of a uniform imaginary black enclosure where an occupant would exchange the same amount of heat by radiation plus convection as in the actual nonuniform environment. An adaptive model, recognizing the influence of outdoor climate on indoor comfort and human adaptation to seasonal temperature variations, determines the appropriate temperature range for comfort evaluation.

3 RESULTS AND ANALYSIS

3.1 Schedule

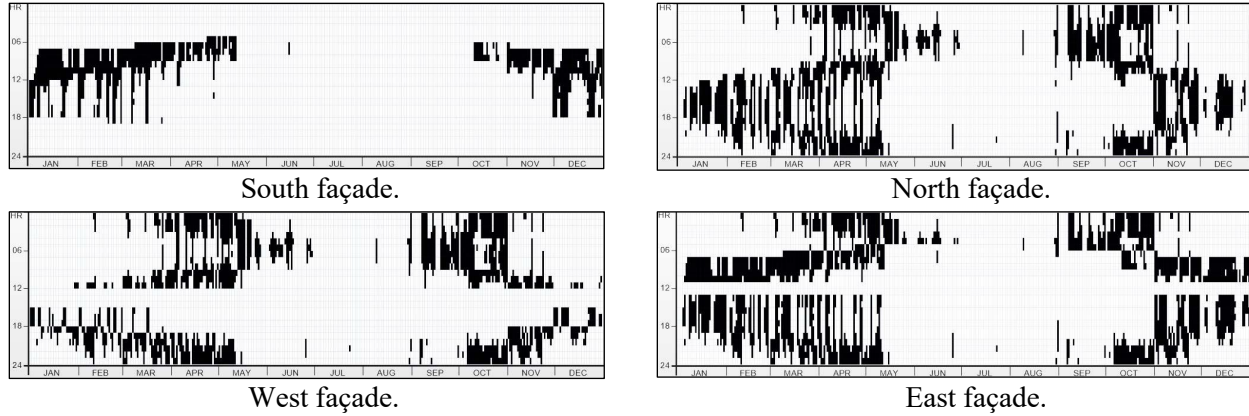


Figure 7: Operation schedule of the facades.

The schedule here comes from the “schedule” concept in Grasshopper meaning the states of an object over the simulation periods, which is normally assigned to objects like shadings, HVAC systems (Heating, Ventilation, and Air Conditioning), and windows. In this case, schedules are assigned to the insulated shutters for their open and closed states. A Honeybee component called “HB Window Construction Shade” is adopted here, through which the thickness, conductivity, transmittance, and reflectance values can all be applied to the shutters. The shutters in the simulation are considered exterior shadings with significant insulation value, and the “HB Fixed Interval Schedule” component is connected to the shading to determine whether the shutters of each façade are applied to the house or not. The schedules of different facades are based on the control logic mentioned in the previous section and shown in Figure 7, the white color indicates the façade is closed and the black color means the façade is open. The schedule patterns are generally based on the control logic and influenced by three temperature thresholds (T_1 , T_2 , T_3 in Figure 3), the schedules presented in Figure 7 correspond to the temperature combination of 5°C, 20°C, and 27°C as an example. In this case, the south façade closes for 7385 hours within 8760 hours per year, the north closes for 7048 hours, the east closes for 6670 hours and the west closes for 7272 hours. It shows that all façades are closed for more or less 7000 hours within a year to achieve a high level of building performance, which means the shutters are supposed to be closed most time of the year to tackle the extreme climate in Phoenix.

3.2 Optimization and Weighting

Once the schedules are set, the faces of the envelope are combined as an “HB Room”, then the ventilation setting (HB Ventilation Control), window setting (HB Window Opening), room loads setting (HB Apply Load Values), and internal mass (HB Internal Mass) are applied to the room and compose as “HB Model”. In this prototype simulation, rooftop water bags and translucent panels on corners are included for better thermal comfort and daylighting results respectively, but rotatable rooftop shutters and skylights are not added because they are not the main objectives of this study but could complicate the simulation process and prolong the simulation time in the initial testing. To evaluate the building performance, the Honeybee Model is transformed into the Open Studio Model, through which the room comfort results can be read out (HB Read Room Comfort Result). The main analytic targets include energy use intensity (EUI), predicted mean vote (PMV) comfort, operative temperature, and useful daylight illuminance (UDI). Adaptive comfort can be an essential value to be analyzed as well, but it is excluded from result evaluation as it failed to run after the authors’ several attempts, the problem may come from the applied schedules since it works well in static simulation models. Once the data of those results are shown, the next step is to optimize the results.

Colibri in Grasshopper can iterate through given combinations of temperature threshold inputs and compile the data of building performance into a CSV file, then Design Explorer is used to visualize the optimization result. Colibri and Design Explorer are iteration tools developed by Thornton Tomasetti for the simulation result analysis and visualization in Grasshopper. Building performance values are multiplied into a new value called “fitness”, and the ranking of optimization results is solely based on the “fitness” value. It should be mentioned that different building performance values can be unequally important from the authors’ perspective, so the “fitness” value is not simply the multiplication of all performance data, more important values are supposed to have higher weights. In this study, thermal comfort is prioritized as the most important factor since the main obstacle this project tries to tackle is extreme heat, and the control logic is mostly based on temperature. UDI is the secondary factor because the interior daylighting condition is already acceptable with skylights and translucent panels even if the shutters are all closed, but soft daylight is always desirable and may improve the living quality. The least important factor is EUI because the electricity generated by rooftop PV panels can exceed the energy demand based on preliminary calculations, and the extra energy cannot be transferred to the grid as the house is isolated. Therefore, the formula of “fitness” calculation is:

$$Fitness = UDI^2 \times PMV\% \times Temp\%^2 \times \frac{1}{EUI}. \quad (1)$$

Thermal comfort is represented by “PMV comfort% x Temp comfort% ^2”, which has the highest weight, “average UDI ^2” stands for daylighting, and “1/EUI” reflects the energy use. The weighting decision here is based on the authors’ understanding of this architectural project in the context, which can be varied in different environments or residents’ needs. For instance, if the residents have a high demand for natural daylighting and the sky coverage is high most of the days, UDI can be highly weighted. Once the weighting is determined, the optimized results can be observed through iterations. Following a series of preliminary tests aimed at identifying the optimal ranges for temperature thresholds T_1 , T_2 , and T_3 , a comprehensive iteration process was undertaken for optimization. The result of this optimization, as depicted in Figure 9, presents two distinct sets of temperature thresholds, each aligning with specific performance criteria.

The first set, exhibiting the highest fitness value of 17.04, establishes temperature thresholds at 5 degrees (T_1), 19 degrees (T_2), and 24 degrees (T_3). This configuration is notable for its superior energy consumption performance, quantified as 1/EUI (Energy Use Intensity). Conversely, the second set, with a fitness value closely trailing 16.89, proposes thresholds of 5 degrees (T_1), 20 degrees (T_2), and 27 degrees (T_3). While its energy consumption performance does not match the first set, it excels in Temp Comfort – an indicator of the duration throughout the year during which occupants experience optimal thermal comfort. Upon comparison, it is observed that while the first group possesses marginally higher fitness, the second group achieves 99.125% of this fitness level, indicating negligible disparity. However, the first group’s predominant advantage in energy consumption performance is somewhat diminished by the relatively

minor weight this factor holds in the overall evaluation criteria. Furthermore, considering the climatic context of Arizona, the 24 degrees (T_3) in the first group appears somewhat restrictive. In contrast, the second group, with its emphasis on Temp Comfort and a higher T_3 threshold, offers a more balanced and occupant-centric approach.

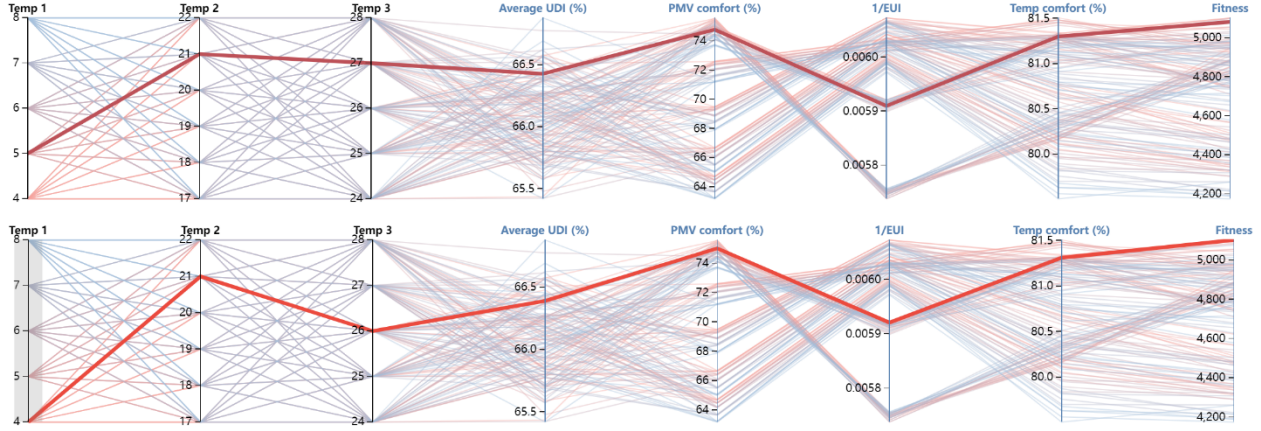


Figure 8: Iterations of optimization results (Temp unit: °C, UDI unit: %).

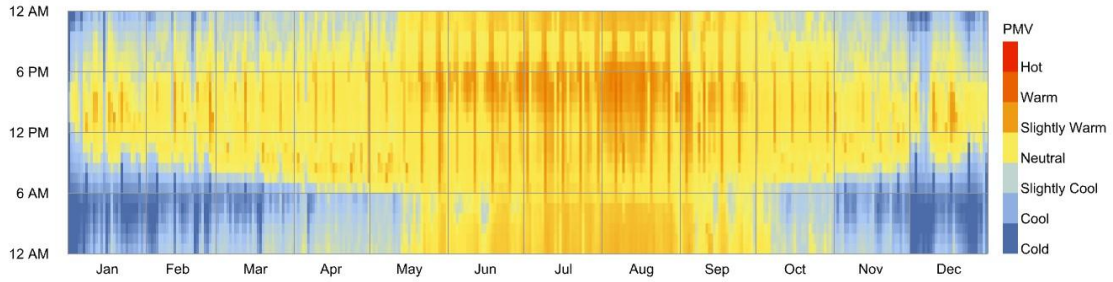


Figure 9: Hourly chart of predicted mean vote (PMV) throughout the year.

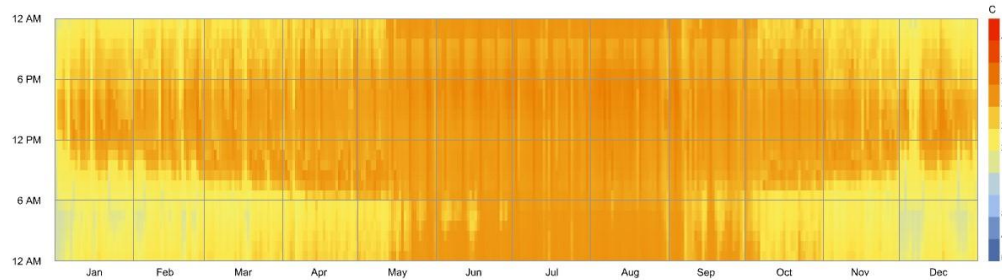


Figure 10: Hourly operative temperature throughout the year.



Figure 11: Annually average useful daylight illuminance (UDI) on the floor area.

The building performance results presented here are based on temperature thresholds of 5°C, 20°C, and 27°C for example. Figure 9 shows the hourly PMV values along the year, and the comfort band between slightly warm and slightly cool takes up 96% time of the year. Figure 10 indicates the operative temperature with the highest of 31.2°C and the lowest of 14.7°C. If the range of 20 to 28°C is defined as a comfortable operative temperature range, 81.7% time of year is comfortable. Figure 11 shows the average UDI on the floor area without the central core as it is a closed space without daylighting. The annual average UDI of the house at a height of 0.8m (sitting height) is 47.0%. Those results are pieces of evidence that the double façade with the insulated shutter system and inner glazing can contribute to high building performance in the context of extreme heat and under the control logic presented above.

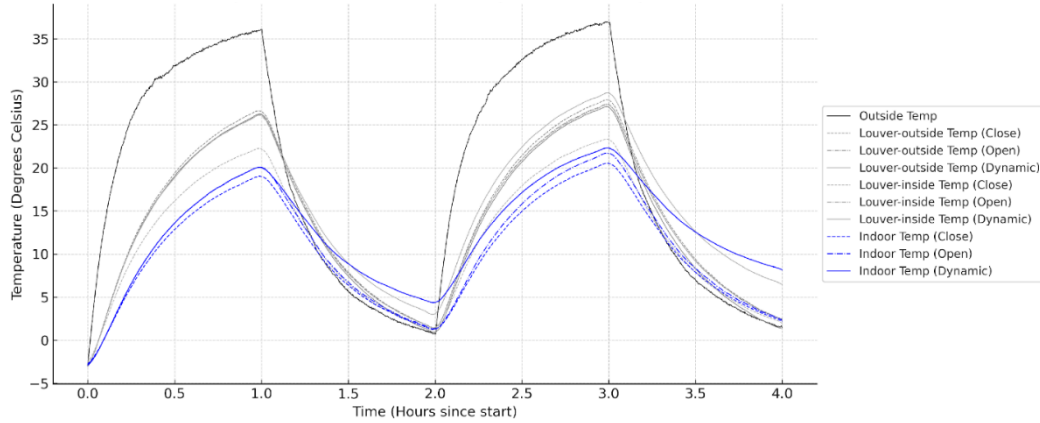


Figure 12: Temperature dynamics in hotbox experiment.

In physical prototype simulations, especially after conducting experiments with three different power levels, the hotbox experiment exposed to medium power is selected to illustrate temperature changes at various sensor locations. According to Figure 12, in each trial, Louver-outside, representing the external part, consistently shows a trend closest to Outside Temp. Generally, Louver-inside aligns closely with Indoor temperature trends, except when louvers are fully open - in this state, both inside and outside of the louvers are equally exposed to the external environment. The critical observation is on indoor temperature. With the louvers completely closed, even though the sealed indoor environment prevents excessive heat exchange and avoids extreme temperatures, it also hinders heating in the scaled model, leading to a relatively low maximum indoor temperature. Therefore, at the experiment's end, the indoor temperature does not significantly exceed that of the fully open state. This underlines the dynamic envelope's advantage: when outdoor temperatures rise necessitating heating, an open façade enables quicker warming. Conversely, when outdoor temperatures drop requiring heat retention, a closed façade substantially stabilizes indoor temperature. In stable experimental conditions, the temperature in fully open or closed states fluctuates around 3 degrees, whereas the dynamic envelope maintains temperature at about 7.5 degrees.

Based on the adaptive model previously described, each environmental temperature corresponds to a specific comfortable temperature range. Utilizing the PMV (Predicted Mean Vote) approach, common in optimization, calculating the proportion of time that the indoor temperature remains within the comfort temperature range relative to the total experiment duration yields a comfort ratio. This metric effectively reflects the impact of the dynamic envelope on human perception and comfort within buildings. As indicated in Figure 13, the comfort ratios for both open and closed louver configurations are similar. However, the comfort ratio for the dynamic envelope is significantly higher, with an improvement of 20% and 25% over the open and closed louvers configurations, respectively. In the experimental design, three distinct power levels of heat sources - low, medium, and high - correspond to increasingly higher temperatures. Observations indicate that the enhancement in comfort ratio afforded by the dynamic envelope amplifies with the intensifying heat source. This finding underscores the dynamic envelope's efficacy in extreme conditions, such as intense heat, which is particularly relevant given the Phoenix climate

used as the project's reference. The dynamic envelope proves highly effective in managing the climatic conditions prevalent in Phoenix.

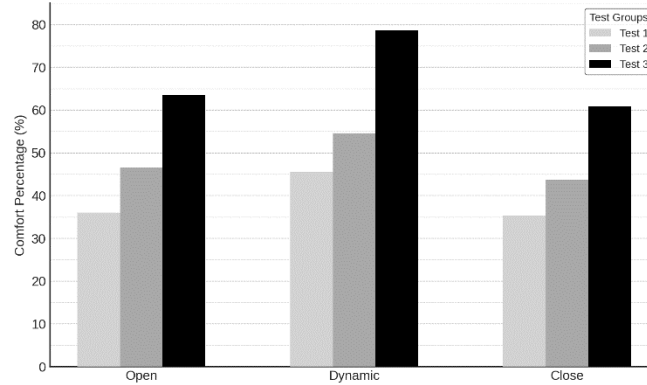


Figure 13: Comparing Comfort Ratio by States.

3.3 Potential Future Works on Model Predictive Control (MPC)

Model Predictive Control, similar to human decision-making processes, is employed to preemptively manage facade changes based on the best-predicted outcomes within a specified timeframe [11]. Given the upper and lower limits of the comfort temperature range, proactive adjustments to the louvers are essential. As temperatures rise and approach the upper limit of the comfort range, louvers can be closed to slow down the temperature increase. A similar approach is applied when temperatures fall. By preemptively adjusting indoor openness or closure, MPC may stabilize the overall temperature, thereby enhancing indoor comfort. While MPC is not applied algorithmically in this research due to time and data restraints, the potential for future application is already proven. Through several experiments involving adjustments to temperature thresholds, despite inherent uncontrollability and potential for unexpected errors in manual experimentation, the comfort results may exceed those of the original mechanically activated dynamic model by nearly 20%.

4 CONCLUSION

This study proposed an insulated shutter system that functions as a secondary envelope of a small desert house. The simulation results prove the hypothesis that the shutter system can enhance building performance significantly and conserve energy. For instance, the operational temperature of the house is between 14.7 and 31.2°C, while the dry-bulb temperatures of Phoenix can be lower than 0°C and higher than 45°C within a year. Additionally, prototype experiments in Hotbox have vividly demonstrated the advantages of using a dynamic envelope compared to a static one (whether open or closed). The dynamic envelope yields a markedly higher comfort ratio, outperforming open and closed louvers configurations by 20% and 25%, respectively. Further, its performance enhancement correlates positively with heat source intensification, demonstrating a 30% efficacy boost in extreme weather conditions. Meanwhile, the distinguished features of the intelligent operation are demonstrated and highlighted. Specifically, the timely opening and closing of the shutter system can regulate indoor temperature dynamics while satisfying daylight and natural ventilation requirements.

Distinct from other typical ABEs relying on measuring the azimuth and altitude angle to determine the change of facade form [11], or systems controlled by temperature and incident solar radiation [5], the control logic of the proposed insulated shutter system is based on solar position (time), outdoor and indoor temperature. This approach yields a straightforward, systematic, and architecturally integrated variant of movable window insulation, which is very different from the precedent systems attached to a single window and embedded with complicated control logic established on thermodynamic laws [5]. Besides, the optimization process provides users the flexibility to maximize their favored indoor comfort aspects or energy conservation, while previous studies tend to focus on energy alone. Furthermore, manual control

options are incorporated to accommodate residents' specific needs or unexpected conditions. To conclude, it is a user-centric design that is integrated into the architecture rather than an accessorial product.

The physical prototype used temperature and light conditions to control the shutters' operation, which can effectively regulate each facade's status. The temperature and sunlight dynamics are predominantly influenced by solar position or time, with real-time testing to account for anomalies such as cloudy days. This control is further refined by using real-time measured light intensity as a climatic correction factor, enabling a comprehensive consideration of various determinants in managing the building's environment. The uniqueness of this design indeed makes it hard to implement into existing buildings or typical architectural projects since they normally do not have fully open facades. However, this prototype can be used for prefabricated housing projects in extreme climates. Considering the system's efficacy in extreme climates, like those in Phoenix, it is expected that for other environmental conditions, the adaptation merely requires simulation to explore new optimized temperature thresholds. This adaptability positions the facade system for widespread application in diverse climatic scenarios. Moreover, the research's deployment of sensors, control systems, materials, and facade components advocates for their efficient and expedient customization, offering promising prospects as a reference for modularization and prefabrication in environmental building design. The investigation in this case study, thus, not only contributes to the field of adaptive building envelopes influenced by light and temperature but also paves the way for responsive, human-centric architectural solutions.

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